MTH201 - Multivariable Calculus (Quiz #3)

Question # 3 of 10 (Start time: 01:58:17 PM, 28 June 2025)

$$\int_0^{\frac{\pi}{2}} \sin^2\theta \; d\theta \; = \; - - - - - - - - -$$

Select the correct option

◉	$\frac{1}{2} \cdot \frac{\pi}{2}$
○	$\frac{2}{3}$
○	$\frac{1}{2}$
○	$\frac{1}{3} \cdot \frac{\pi}{2}$

MTH201 - Multivariable Calculus (Quiz #3)

Question # 1 of 10 (Start time: 02:04:47 PM, 28 June 2025)

$$\int_0^{\frac{\pi}{2}} \sin^2\theta \; d\theta \; = \; - - - - - - - - -$$

Select the correct option

○	$\frac{1}{2}$
○	$\frac{2}{3}$
○	$\frac{1}{3} \cdot \frac{\pi}{2}$
●	$\frac{1}{2} \cdot \frac{\pi}{2}$

MTH201 - Multivariable Calculus (Quiz #3)

Question # 2 of 10 (Start time: 02:05:14 PM, 28 June 2025)

The differential equation $dz = x^2\,dx + y^2\,dy$ is an exact differential equation.

Select the correct option

◉	False
○	True

MTH201 - Multivariable Calculus (Quiz #3)

Question # 8 of 10 (Start time: 02:00:57 PM, 28 June 2025)

One of the line integral properties is $\int_{AB} Fds = -\int_{BA} Fds$

Select the correct option

◉	True
○	False

Click

MTH201 - Multivariable Calculus (Quiz #3)

Question # 1 of 10 (Start time: 01:57:49 PM, 28 June 2025)

Integration along two distinct paths joining the same two end points ..... give the same results.

Select the correct option

- ○ always
- ◉ not necessarily

MTH201 - Multivariable Calculus (Quiz #3)

Question # 7 of 10 (Start time: 02:00:37 PM, 28 June 2025)

If $\vec{r}(t) = t\,\hat{i} + 2t\,\hat{j}$, then $\frac{d}{dt}[\vec{r}(t)] =$ _____.

Select the correct option

- ○ $t\,\hat{i} + 2\hat{j}$
- ○ $\hat{i} + \hat{j}$
- ○ $\hat{i} + 2t\,\hat{j}$
- ◉ $\hat{i} + 2\hat{j}$

MTH201 - Multivariable Calculus (Quiz #3)

Question # 10 of 10 (Start time: 02:08:15 PM, 28 June 2025)

In the integration of polar coordinates $dxdy$ is replaced by

Select the correct option

○	rdr
●	$rdrd\theta$
○	$d\theta$
○	$drd\theta$

MTH201 - Multivariable Calculus (Quiz #3)

Question # 2 of 10 (Start time: 01:58:02 PM, 28 June 2025)

The vector $\vec{r}$ is continuous at t_0 if _____.

Select the correct option

●	(d) All (a), (b) and (c).
○	(c) $\lim_{t \to t_0} \vec{r}(t) = \vec{r}(t_0)$.
○	(b) $\lim_{t \to t_0} \vec{r}(t)$ exist.
○	(a) $\vec{r}(t_0)$ is defined.

MTH201 - Multivariable Calculus (Quiz #3)

Question # 3 of 10 (Start time: 02:05:25 PM, 28 June 2025)

The graph of $r = (1 + t)i + (-2 + 3t)j - 4tk$ is the

Select the correct option

○	line that passes through the point (1, 3, -4)
●	line that passes through the point (1, -2, 0).
○	line that passes through the point (2, 1, -4)
○	None of these

MTH201 - Multivariable Calculus (Quiz #3)

Question # 4 of 10 (Start time: 01:58:53 PM, 28 June 2025)

In 2D - space the parametric equations $x = x(t), y = y(t)$ can be expressed in single vector equationas

Select the correct option

○	$\vec{r}(t) = x(t)j + y(t)i$
○	$\vec{r}(t) = x(t) + y(t)$
○	$\vec{r}(t) = x(t) - y(t)$
●	$\vec{r}(t) = x(t)i + y(t)j$

Click

Question # 5 of 10 (Start time: 02:06:10 PM, 28 June 2025) Total Marks: 1

The differential equation, $dz = (2x^2 - 2xy + 3)\,dx + (6y^2 - 2x^2 + 1)\,dy$, is an exact differential equation.

Select the correct option

Reload Math Equations

- (●) False
- ( ) True

MTH201 - Multivariable Calculus (Quiz #3)

Question # 9 of 10 (Start time: 02:07:39 PM, 28 June 2025)

In polar coordinates, $\iint_R f(x,\ y)\,dx\,dy\ =\ ------$

Select the correct option

- ( ) $\iint_G f(r\cos\theta,\ r\sin\theta)\,rdr\,d\theta$
- ( ) $\iint_G f(r\cos\theta,\ r\sin\theta)\,dr\,d\theta$
- ( ) $\iint_G f(r\sin\theta,\ r\cos\theta)\,rdr\,d\theta$
- (●) $\int_G f(r\cos\theta,\ r\sin\theta)\,rdr\,d\theta$

MTH201 - Multivariable Calculus (Quiz #3)

Question # 8 of 10 (Start time: 02:07:01 PM, 28 June 2025)

Given the integral $\iint_R f(x,y)dxdy$, can be expressed in ploar coordinates as. . . . . ,
where $a \leqslant \theta \leqslant b$ and $c \leqslant r \leqslant d$.

Select the correct option

○	$\int_a^b \int_c^d f(r,\theta) r d\theta dr$
◉	$\int_a^b \int_c^d f(r,\theta) r dr d\theta$
○	$\int_a^b \int_c^d f(r,\theta) dr d\theta$
○	$\int_a^c \int_b^d f(r,\theta) r d\theta dr$

C

MTH201 - Multivariable Calculus (Quiz #3)

Question # 6 of 10 (Start time: 01:59:59 PM, 28 June 2025)

If $x'(t), y'(t)$ and $z'(t)$ are continuous for $a \leqslant t \leqslant b$, then the given parametric equations are

Select the correct option

◉	$x = x(t), y = y(t), z = z(t)\ ; (a \leqslant t \leqslant b)$
○	$x = x(t) + y(t) + z(t)\ ; (a \leqslant t \leqslant b)$
○	$x = x'(t), y = y'(t), z = z'(t)\ ; (a \leqslant t \leqslant b)$
○	$x = x'(t) + y'(t) + z'(t)\ ; (a \leqslant t \leqslant b)$

C

MTH201 - Multivariable Calculus (Quiz #3) Quiz Start Time: 02:04 PM, 28 June 2025

Question # 4 of 10 (Start time: 02:05:42 PM, 28 June 2025) Total Marks: 1

Equation of the tangent line of vector valued function $r(t)$ at $r(t_0)$ is

Select the correct option

Reload Math Equations

- $r = r(t_0)$
- $r = r(t_0) + r'(t_0)$
- $r = r(t_0) + t$
- (selected) $r = r(t_0) + tr'(t_0)$

Click to Save Answer & Move to Next Question

MTH201 - Multivariable Calculus (Quiz #3) Quiz Start Time: 01:57 PM, 28 June 2025

Question # 10 of 10 (Start time: 02:01:44 PM, 28 June 2025) Total Marks: 1

If $\vec{r}(t) = t^2\,\hat{i} + t\,\hat{j} + 3\,t\,\hat{k}$, then $\frac{d}{dt}[\vec{r}(t)] =$ _____.

Select the correct option

Reload Math Equations

- (selected) $2\,t\,\hat{i} + \hat{j} + 3\,\hat{k}$
- $\hat{i} + t\,\hat{j} + 3\,\hat{k}$
- $\hat{i} + \hat{j} + \hat{k}$
- $2\,t\,\hat{i} + t\,\hat{j} + 3\,\hat{k}$

Click to Save Answer & Move to Next Question

MTH201 - Multivariable Calculus (Quiz #3) Quiz Start Time: 02:04 PM, 28 June 2025

Question # 7 of 10 (Start time: 02:06:42 PM, 28 June 2025) Total Marks: 1

In 2D - space the parametric equations $x = x(t), y = y(t)$ can be expressed in single vector equationas

Select the correct option

Reload Math Equations

- $\vec{r}(t) = x(t)j + y(t)i$
- $\vec{r}(t) = x(t) + y(t)$
- $\vec{r}(t) = x(t) - y(t)$
- (selected) $\vec{r}(t) = x(t)i + y(t)j$

Click to Save Answer & Move to Next Question

MTH201 - Multivariable Calculus (Quiz #3) Quiz Start Time: 02:04 PM, 28 June 2025

Question # 6 of 10 (Start time: 02:06:21 PM, 28 June 2025) Total Marks: 1

For a vector valued function $\vec{r}(t) = 2t\,i + (t-1)\,j$ then lenght of $\vec{r}(t)$

Select the correct option

Reload Math Equations

- $\|r(t)\| = \sqrt{t^2 + (t-1)^2}$
- $\|r(t)\| = \sqrt{2t^2 + (t-1)^2}$
- (selected) $\|r(t)\| = \sqrt{4t^2 + (t-1)^2}$
- $\|r(t)\| = \sqrt{4t^2 - (t-1)^2}$

Click to Save Answer & Move to Next Question

MTH201 - Multivariable Calculus (Quiz #3) Quiz Start Time: 01:57 PM, 28 June 2025

Question # 9 of 10 (Start time: 02:01:12 PM, 28 June 2025) Total Marks: 1

The arc length of the curver$(t) = t^3 i + tj + \frac{1}{2}\sqrt{6}t^2 k; 0 \leqslant t \leqslant 2$, can be written as

Select the correct option

Reload Math Equations

- $L = \int_0^2 \sqrt{9t^4 + 1 - 3t^2}\,dt$
- $L = \int_0^2 \sqrt{9t^4 + 1 - 6t^2}\,dt$
- (selected) $L = \int_0^2 \sqrt{9t^4 + 1 + 6t^2}\,dt$
- $L = \int_0^2 \sqrt{9t^4 + 1 + 3t^2}\,dt$

Click to Save Answer & Move to Next Question

Question # 5 of 10 (Start time: 01:59:18 PM, 28 June 2025) Total Marks: 1

The arc length of the portation of the circular helix where $(dx/dt) = -\sin t, (dy/dt) = \cos t$ and$(dz/dt) = 1$and $0 \leqslant t \leqslant \pi$, then the arc lenght is

Select the correct option

Reload Math Equations

- $L = \int_0^{\pi} \sqrt{2}dy$
- $L = \int_0^{\pi} -\sqrt{2}dt$
- $L = \int_0^{\pi} \sqrt{2}dx$
- $L = \int_0^{\pi} \sqrt{2}dt$

Click to Save Answer & Move to Next Question